

Initiating Search

November 10, 2025, 9:59 PM

• Search:

CAS Lexicon Search:

Knorr pyrrole synthesis - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (198)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (669)	 Reactions	View Results
Filtered By:		
Document Type:	Journal	

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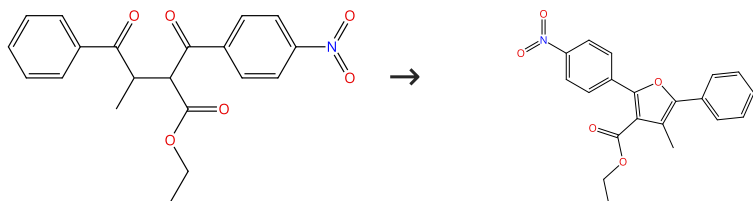
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-571-CAS-7870714

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

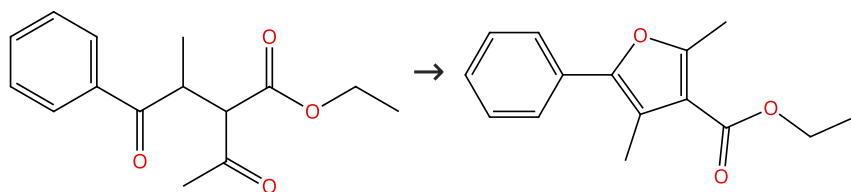
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 4 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (3)

 Suppliers (2)

31-571-CAS-3592812

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

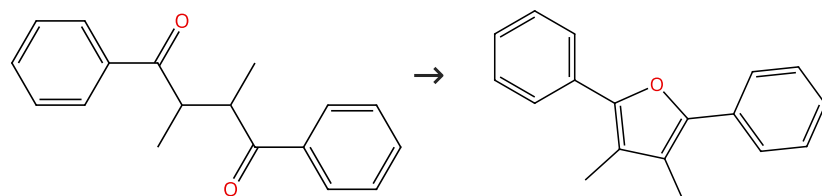
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 3 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (2)

 Suppliers (5)

31-571-CAS-6075761

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

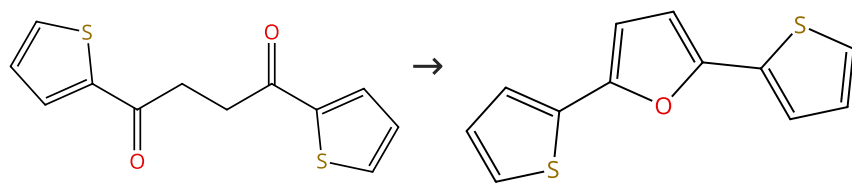
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (8)

31-571-CAS-14925862

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

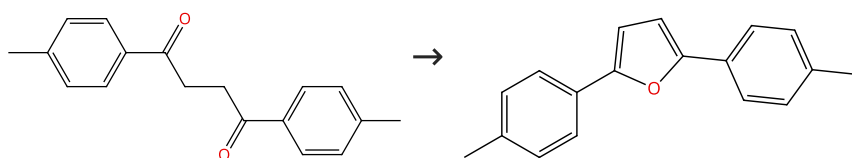
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 3 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

Suppliers (4)

31-571-CAS-5797794

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

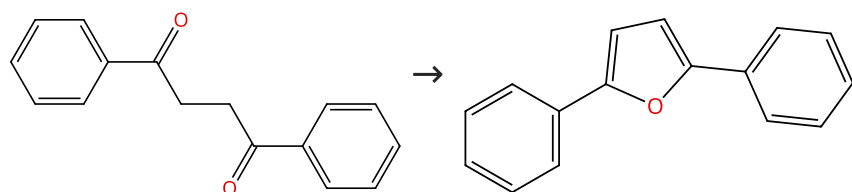
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (60)

31-571-CAS-3641828

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

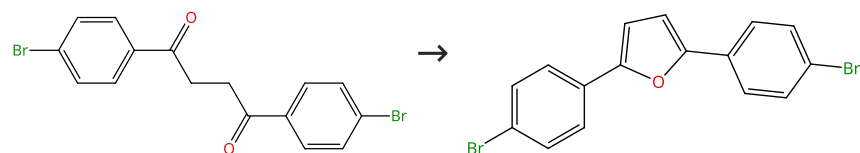
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 7 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (14)

Suppliers (12)

31-571-CAS-329352

Steps: 1 Yield: 100%

N-Substituted pyrrole synthesis by Paal- Knorr condensation using recyclable cationic exchange resin in water

1.1 **Catalysts:** Hydrochloric acid
Solvents: Water; 24 h, reflux

By: He, Yan-Hong; et al

Journal of Heterocyclic Chemistry (2010), 47(2), 486-489.

31-571-CAS-10049121

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

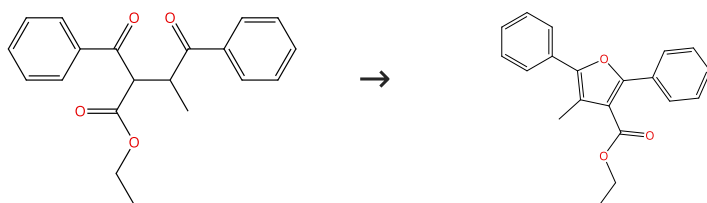
1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

Synthetic Communications (2010), 40(3), 370-377.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (4)

31-571-CAS-5744596

Steps: 1 Yield: 100%

Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis

1.1 **Catalysts:** 1-Butyl-3-methylimidazolium hydrogen sulfate
Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 4 h, 150 °C; 150 °C → rt

By: Wang, Gangqiang; et al

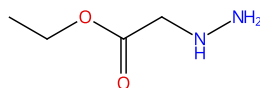
Synthetic Communications (2010), 40(3), 370-377.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%

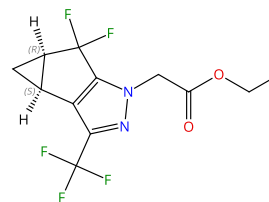
Multi-component
 structure image
 available in CAS
 SciFinder

+



• HCl

→



Absolute stereochemistry shown

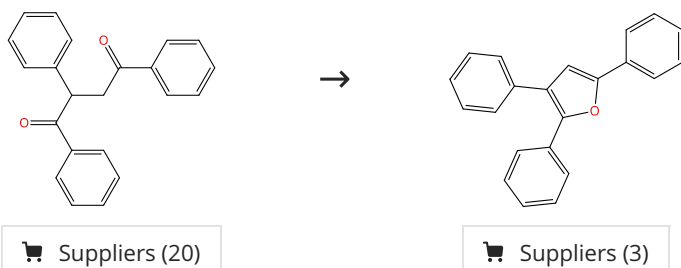
Suppliers (72)

Suppliers (2)

31-614-CAS-41421477 Steps: 1 Yield: 100% 1.1 Reagents: Acetyl chloride, Lithium chloride Solvents: Ethanol; 18.5 h, 55 °C 1.2 Reagents: 1,3-Dibromo-5,5-dimethylhydantoin, Pyridine, hydrofluoride (1:1) Solvents: Dichloromethane; 40 min, -13 -- -11 °C; 4 h, -11 - 9 °C 1.3 Reagents: Potassium hydroxide, Sodium pyrosulfite Solvents: Water; 0 °C; 40 min, 20 °C Experimental Protocols	Synthetic Development and Scale-Up of a Complex Pyrazole Fragment of Lenacapavir By: Kadunce, Nathaniel; et al Organic Process Research & Development (2024), 28(8), 3368-3381.
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Scheme 10 (1 Reaction)

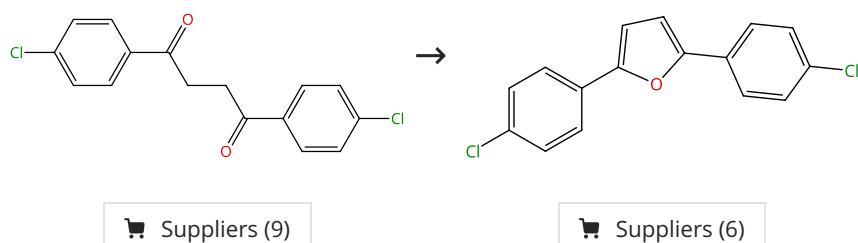
Steps: 1 Yield: 100%



31-571-CAS-3943300 Steps: 1 Yield: 100% 1.1 Catalysts: 1-Butyl-3-methylimidazolium hydrogen sulfate Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt	Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis By: Wang, Gangqiang; et al Synthetic Communications (2010), 40(3), 370-377.
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Scheme 11 (1 Reaction)

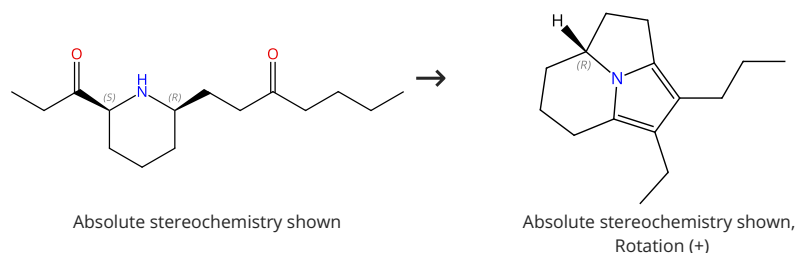
Steps: 1 Yield: 100%



31-571-CAS-7921970 Steps: 1 Yield: 100% 1.1 Catalysts: 1-Butyl-3-methylimidazolium hydrogen sulfate Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 2 h, 150 °C; 150 °C → rt	Ionic liquid as catalyst and reaction medium. A simple and efficient procedure for Paal-Knorr furan synthesis By: Wang, Gangqiang; et al Synthetic Communications (2010), 40(3), 370-377.
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Scheme 12 (1 Reaction)

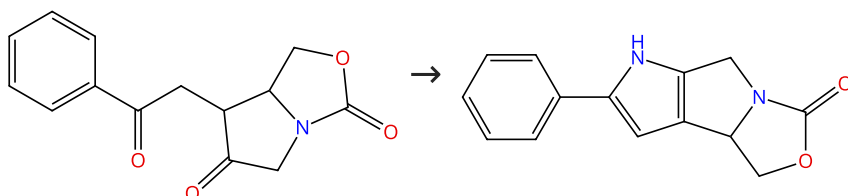
Steps: 1 Yield: 99%



31-309-CAS-13799040	Steps: 1 Yield: 99%	Synthetic and Theoretical Investigations of Myrmicarín Biosynthesis By: Snyder, Scott A.; et al Angewandte Chemie, International Edition (2010), 49(50), 9693-9698, S9693/1-S9693/86.
1.1 Reagents: Methanol Solvents: Methanol; 1 h, 23 °C		
Experimental Protocols		

Scheme 13 (1 Reaction)

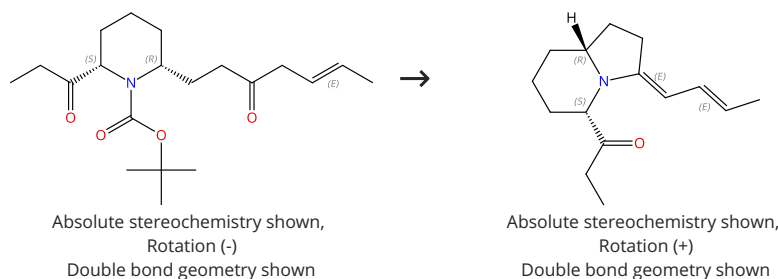
Steps: 1 Yield: 99%



31-574-CAS-8785889	Steps: 1 Yield: 99%	Enone-alkyne reductive coupling: a versatile entry to substituted pyrroles By: Thompson, Benjamin B.; et al Organic Letters (2011), 13(13), 3289-3291.
1.1 Reagents: Acetic acid, Ammonium acetate Solvents: Tetrahydrofuran; 15 min, 170 °C		
Experimental Protocols		

Scheme 14 (1 Reaction)

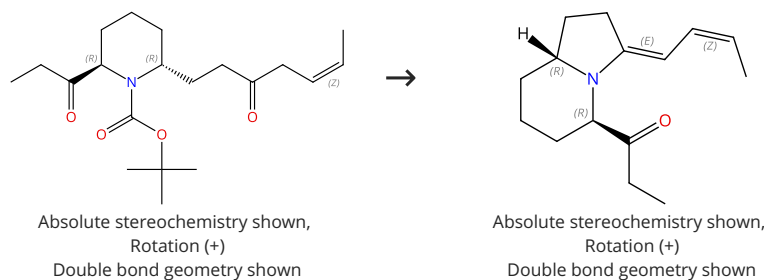
Steps: 1 Yield: 99%



31-574-CAS-11332540	Steps: 1 Yield: 99%	Synthetic and Theoretical Investigations of Myrmicarín Biosynthesis By: Snyder, Scott A.; et al Angewandte Chemie, International Edition (2010), 49(50), 9693-9698, S9693/1-S9693/86.
1.1 Reagents: Trifluoroacetic acid Solvents: Dichloromethane, Water; 0 °C; 1 h, 0 °C		
1.2 Reagents: Potassium carbonate, Sodium hydroxide Solvents: Water; 20 s, cooled		
Experimental Protocols		

Scheme 15 (1 Reaction)

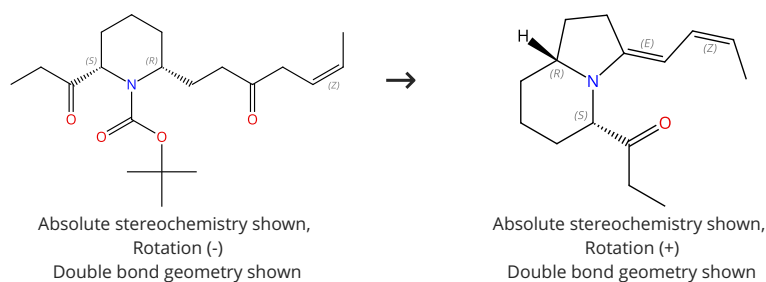
Steps: 1 Yield: 99%



31-574-CAS-2219198	Steps: 1 Yield: 99%	Synthetic and Theoretical Investigations of Myrmicarín Biosynthesis By: Snyder, Scott A.; et al Angewandte Chemie, International Edition (2010), 49(50), 9693-9698, S9693/1-S9693/86.
1.1 Reagents: Trifluoroacetic acid Solvents: Dichloromethane, Water; 0 °C; 1 h, 0 °C		
1.2 Reagents: Potassium carbonate, Sodium hydroxide Solvents: Water; 20 s, cooled		
Experimental Protocols		

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



31-574-CAS-13471850

Steps: 1 Yield: 99%

Synthetic and Theoretical Investigations of Myrmicarins Biosynthesis

By: Snyder, Scott A.; et al

Angewandte Chemie, International Edition (2010), 49(50), 9693-9698, S9693/1-S9693/86.

1.1 Reagents: Trifluoroacetic acid

Solvents: Dichloromethane, Water; 0 °C; 1 h, 0 °C

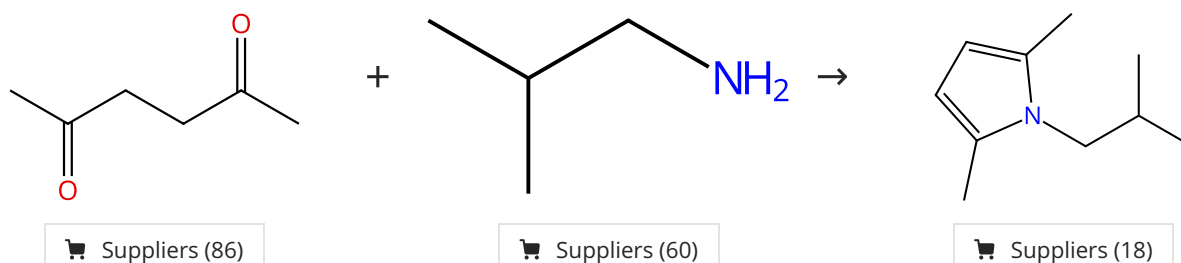
1.2 Reagents: Potassium carbonate, Sodium hydroxide

Solvents: Water; 20 s, cooled

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 99%



31-309-CAS-2150835

Steps: 1 Yield: 99%

Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach

By: Devi, Aarti; et al

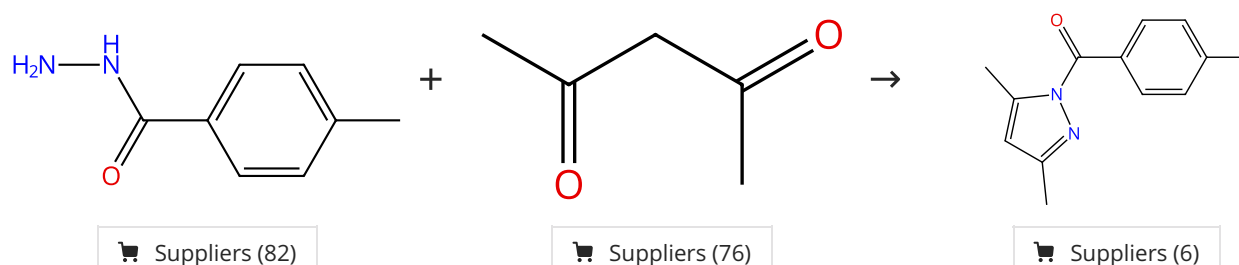
Synthetic Communications (2012), 42(10), 1480-1488.

1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 6 min, rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-24902728

Steps: 1 Yield: 99%

Iodonium salts as efficient iodine(III)-based noncovalent organocatalysts for Knorr-type reactions

By: Yunusova, Sevilya N.; et al

RSC Advances (2021), 11(8), 4574-4583.

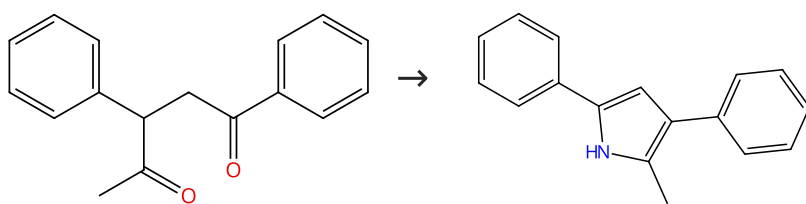
1.1 Catalysts: Dibenziodolium, salt with trifluoromethanesulfonic acid (1:1)

Solvents: Methanol; 3.5 h, 50 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (4)

Suppliers (4)

31-574-CAS-13369722

Steps: 1 Yield: 99%

Enone-alkyne reductive coupling: a versatile entry to substituted pyrroles

1.1 **Reagents:** Acetic acid, Ammonium acetate
Solvents: Tetrahydrofuran; 15 min, 170 °C

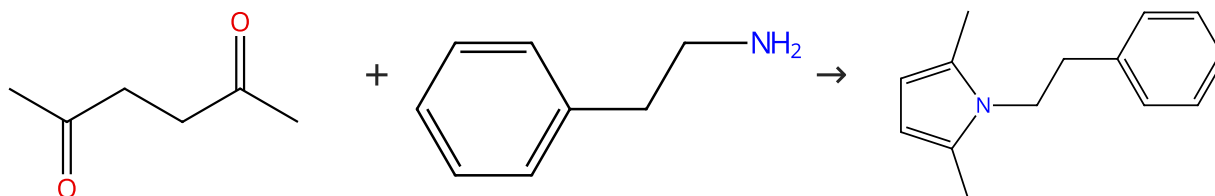
By: Thompson, Benjamin B.; et al

Organic Letters (2011), 13(13), 3289-3291.

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (55)

Suppliers (14)

31-309-CAS-13426226

Steps: 1 Yield: 99%

Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach

1.1 **Catalysts:** Amberlite IR 120 (acidic resin bound); 14 min, rt

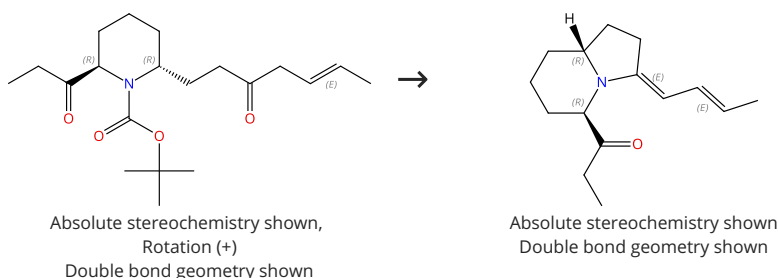
By: Devi, Aarti; et al

Synthetic Communications (2012), 42(10), 1480-1488.

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



31-574-CAS-110743

Steps: 1 Yield: 99%

Synthetic and Theoretical Investigations of Myrmecarin Biosynthesis

1.1 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane, Water; 0 °C; 1 h, 0 °C

1.2 **Reagents:** Potassium carbonate, Sodium hydroxide
Solvents: Water; 20 s, cooled

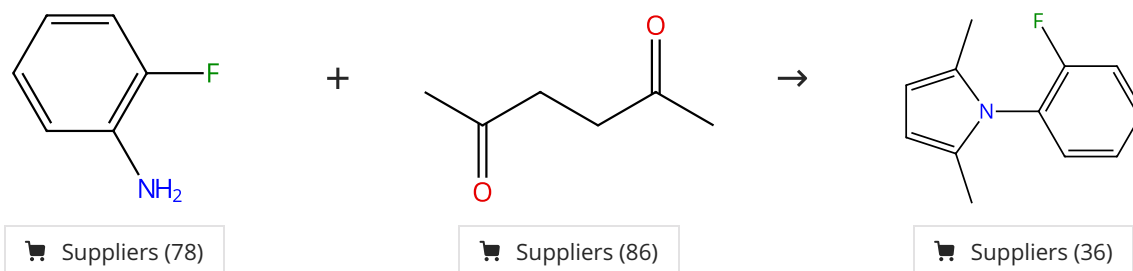
By: Snyder, Scott A.; et al

Angewandte Chemie, International Edition (2010), 49(50), 9693-9698, S9693/1-S9693/86.

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



31-309-CAS-8887673

Steps: 1 Yield: 99%

Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach

1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 30 min, rt

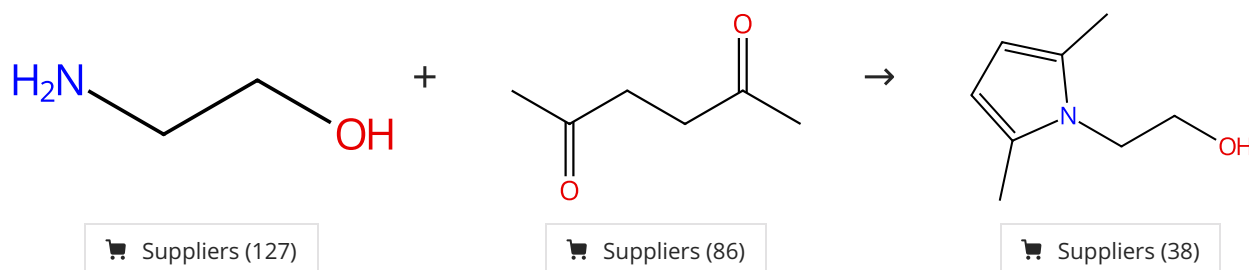
By: Devi, Aarti; et al

Experimental Protocols

Synthetic Communications (2012), 42(10), 1480-1488.

Scheme 23 (2 Reactions)

Steps: 1 Yield: 99%



31-309-CAS-12608696

Steps: 1 Yield: 99%

Multijet Oscillating Disc Millireactor: A Novel Approach for Continuous Flow Organic Synthesis

1.1 18 min, 20 °C

By: Liguori, Lucia; et al

Experimental Protocols

Organic Process Research & Development (2011), 15(5), 997-1009.

31-309-CAS-13574700

Steps: 1 Yield: 99%

Fast Scale-Up Using Microreactors: Pyrrole Synthesis from Micro to Production Scale

1.1 Solvents: Methanol; 100 s, 20 °C

By: Nieuwland, Pieter J.; et al

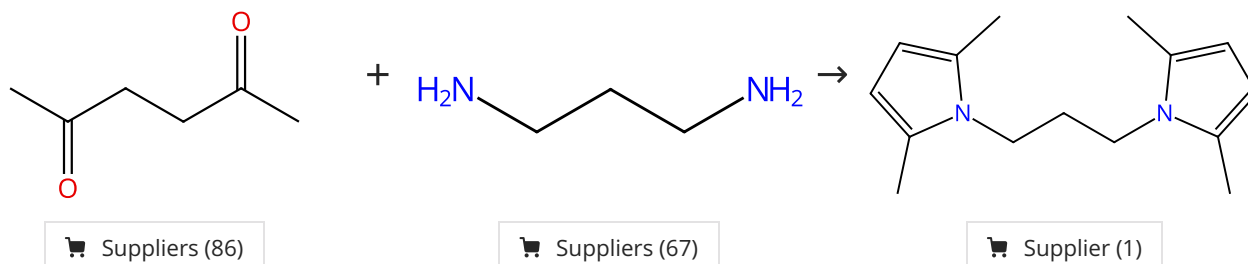
1.2 Reagents: Acetone

Experimental Protocols

Organic Process Research & Development (2011), 15(4), 783-787.

Scheme 24 (1 Reaction)

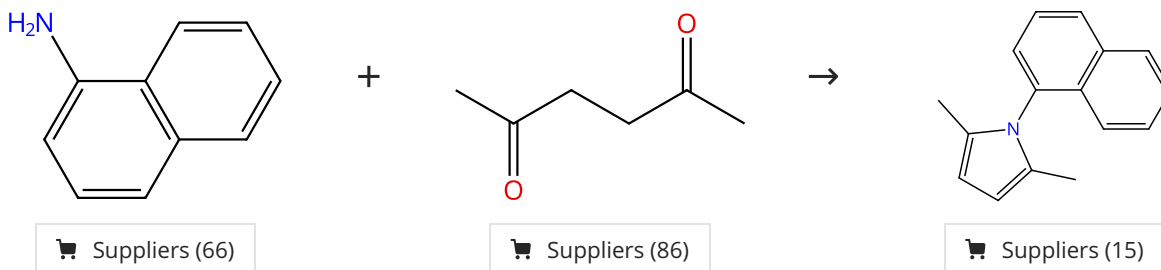
Steps: 1 Yield: 99%



31-309-CAS-4278395	Steps: 1 Yield: 99%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 5 min, rt		By: Devi, Aarti; et al
Experimental Protocols		Synthetic Communications (2012), 42(10), 1480-1488.

Scheme 25 (1 Reaction)

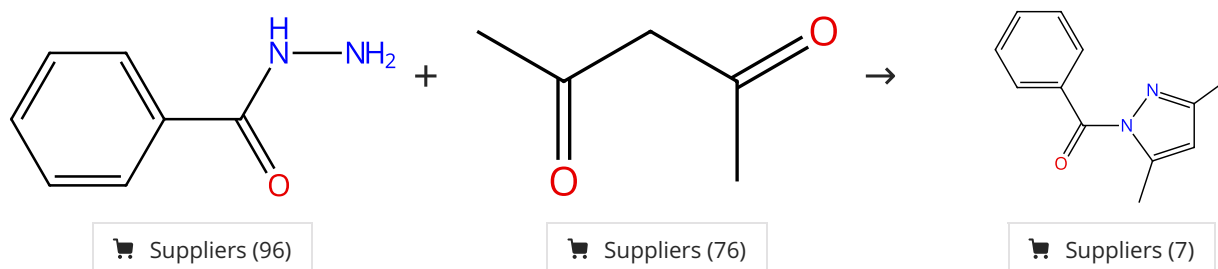
Steps: 1 Yield: 99%



31-309-CAS-8887	Steps: 1 Yield: 99%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 40 min, rt		By: Devi, Aarti; et al
Experimental Protocols		Synthetic Communications (2012), 42(10), 1480-1488.

Scheme 26 (1 Reaction)

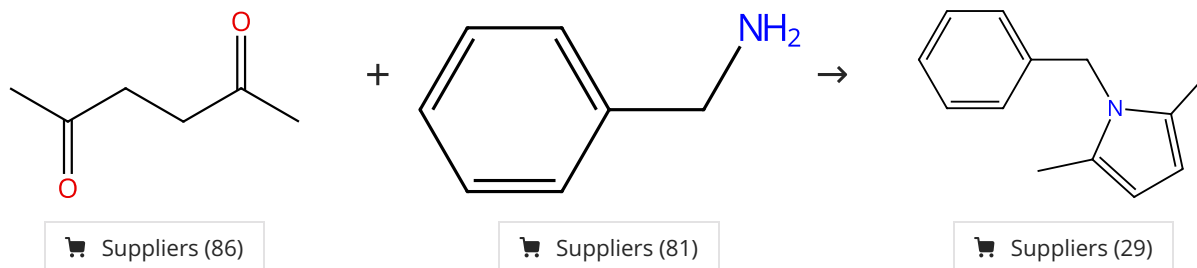
Steps: 1 Yield: 99%



31-614-CAS-24902712	Steps: 1 Yield: 99%	Iodonium salts as efficient iodine(III)-based noncovalent organocatalysts for Knorr-type reactions
1.1 Catalysts: Dibenziodolium, salt with trifluoromethanesulfonic acid (1:1)		By: Yunusova, Sevilya N.; et al
Solvents: Methanol; 3.5 h, 50 °C		RSC Advances (2021), 11(8), 4574-4583.
Experimental Protocols		

Scheme 27 (5 Reactions)

Steps: 1 Yield: 97-99%

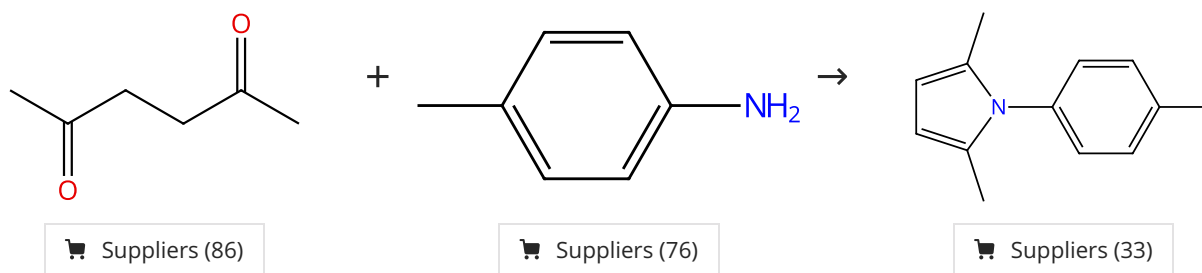


31-309-CAS-11464800	Steps: 1 Yield: 99%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 16 min, rt		By: Devi, Aarti; et al
Experimental Protocols		Synthetic Communications (2012), 42(10), 1480-1488.

31-309-CAS-12256362 Steps: 1 Yield: 99% 1.1 Catalysts: <i>N</i>¹,<i>N</i>¹,<i>N</i>³,<i>N</i>³-Tetrachloro-1,3-benzenedisulfonamide Solvents: Acetonitrile; 2 min, rt	One-pot synthesis of substituted pyrroles with <i>N,N,N',N'</i>-tetrachlorobenzene-1,3-disulfonamide and <i>N,N'</i>-diiodo-<i>N,N'</i>-1,2-ethanediylbis(<i>p</i>-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.
31-309-CAS-3705714 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, Chlorosulfonic acid (silica-supported); 3 min, rt Experimental Protocols	Silica sulfuric acid (SSA) as a solid acid heterogeneous catalyst for one-pot synthesis of substituted pyrroles under solvent-free conditions at room temperature By: Veisi, Hojat Tetrahedron Letters (2010), 51(16), 2109-2114.
31-309-CAS-10125658 Steps: 1 Yield: 98% 1.1 Catalysts: <i>N,N'</i>-1,2-Ethanediylbis[<i>N</i>-iodo-4-methylbenzenesulfonamide] Solvents: Acetonitrile; 10 min, rt	One-pot synthesis of substituted pyrroles with <i>N,N,N',N'</i>-tetrachlorobenzene-1,3-disulfonamide and <i>N,N'</i>-diiodo-<i>N,N'</i>-1,2-ethanediylbis(<i>p</i>-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.
31-309-CAS-4803548 Steps: 1 Yield: 97% 1.1 Catalysts: Zirconium oxychloride; 5 min, rt Experimental Protocols	ZrOCl₂·8H₂O as a highly efficient, eco-friendly and recyclable Lewis acid catalyst for one-pot synthesis of <i>N</i>-substituted pyrroles under solvent-free conditions at room temperature By: Rahmatpour, Ali Applied Organometallic Chemistry (2011), 25(8), 585-590.

Scheme 28 (1 Reaction)

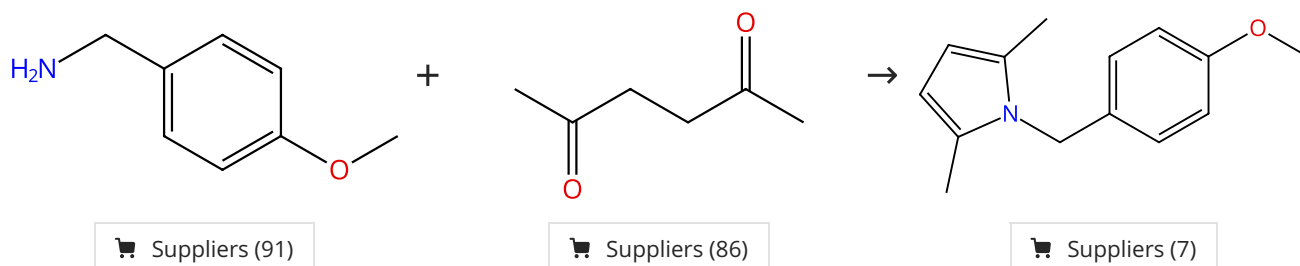
Steps: 1 Yield: 99%



31-309-CAS-304692 Steps: 1 Yield: 99% 1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 20 min, rt Experimental Protocols	Paal-Knorr pyrrole synthesis using recyclable amberlite IR 120 acidic resin, a green approach By: Devi, Aarti; et al Synthetic Communications (2012), 42(10), 1480-1488.
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Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



31-309-CAS-14388494

Steps: 1 Yield: 99%

1.1 Catalysts: *N*¹,*N*¹,*N*³,*N*³-Tetrachloro-1,3-benzenedisulfonamide
 Solvents: Acetonitrile; 3 min, rt

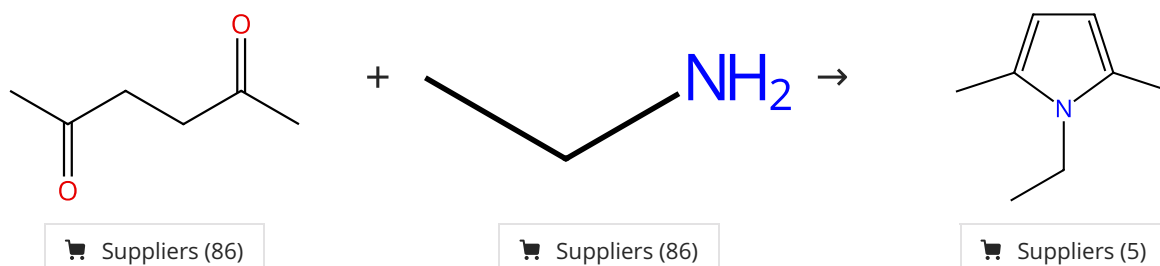
One-pot synthesis of substituted pyrroles with *N*,*N*,*N*',*N*'-tetrachlorobenzene-1,3-disulfonamide and *N*,*N*'-diiodo-*N*,*N*'-1,2-ethanediylbis(*p*-toluenesulfonamide) as novel catalytic reagents

By: Ghorbani-Vaghei, Ramin; et al

South African Journal of Chemistry (2009), 62, 33-38.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



31-309-CAS-7959864

Steps: 1 Yield: 99%

1.1 Solvents: Methanol; 100 s, 20 °C

1.2 Reagents: Acetone

Experimental Protocols

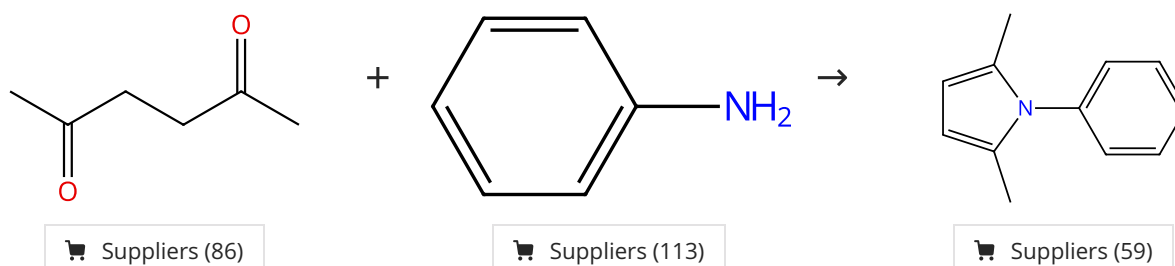
Fast Scale-Up Using Microreactors: Pyrrole Synthesis from Micro to Production Scale

By: Nieuwland, Pieter J.; et al

Organic Process Research & Development (2011), 15(4), 783-787.

Scheme 31 (2 Reactions)

Steps: 1 Yield: 98-99%



31-309-CAS-14899571

Steps: 1 Yield: 99%

1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 18 min, rt

Experimental Protocols

Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach

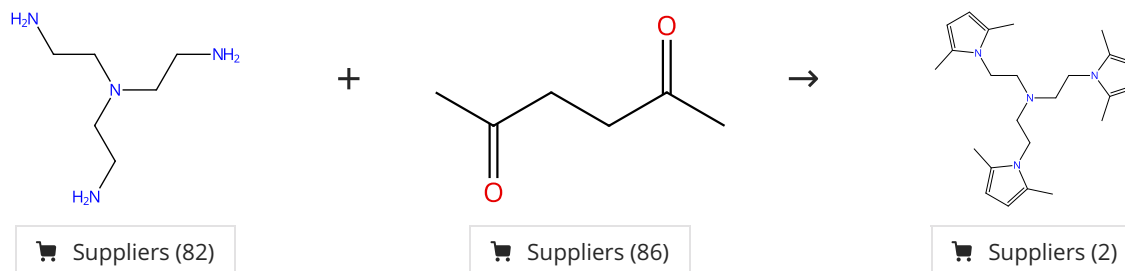
By: Devi, Aarti; et al

Synthetic Communications (2012), 42(10), 1480-1488.

31-309-CAS-7992341	Steps: 1 Yield: 98%	One-pot synthesis of substituted pyrroles with N,N,N',N'-tetrachlorobenzene-1,3-disulfonamide and N,N'-diiodo-N,N'-1,2-ethanediylbis(p-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.
1.1 Catalysts: <i>N</i> ¹ , <i>N</i> ¹ , <i>N</i> ³ , <i>N</i> ³ -Tetrachloro-1,3-benzenedisulfonamide Solvents: Acetonitrile; 5 min, rt		

Scheme 32 (1 Reaction)

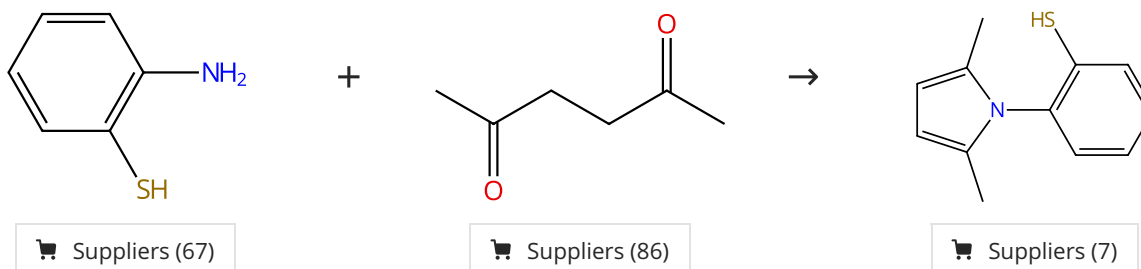
Steps: 1 Yield: 98%



31-309-CAS-10034672	Steps: 1 Yield: 98%	Silica sulfuric acid (SSA) as a solid acid heterogeneous catalyst for one-pot synthesis of substituted pyrroles under solvent-free conditions at room temperature By: Veisi, Hojat Tetrahedron Letters (2010), 51(16), 2109-2114.
1.1 Catalysts: Silica, Chlorosulfonic acid (silica-supported); 5 min, rt Experimental Protocols		

Scheme 33 (1 Reaction)

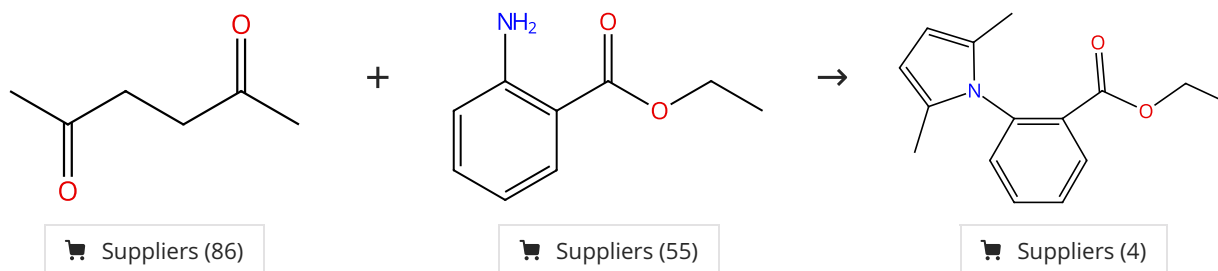
Steps: 1 Yield: 98%



31-309-CAS-15276708	Steps: 1 Yield: 98%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach By: Devi, Aarti; et al Synthetic Communications (2012), 42(10), 1480-1488.
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 36 min, rt Experimental Protocols		

Scheme 34 (1 Reaction)

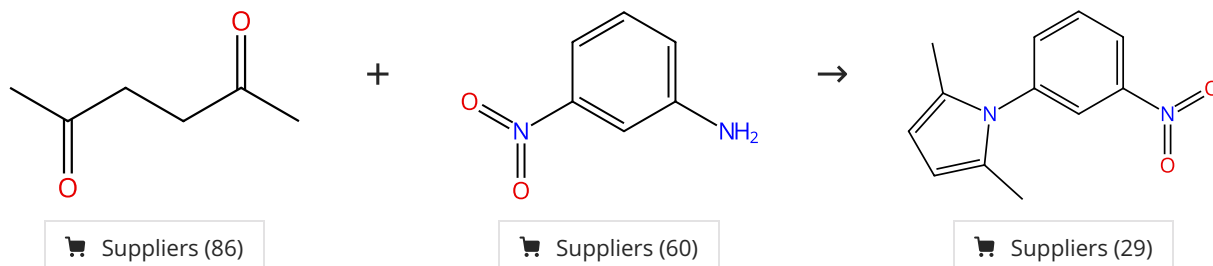
Steps: 1 Yield: 98%



31-309-CAS-9016501	Steps: 1 Yield: 98%	N-(2-Methylinden-4-yl)pyrrole as a heterocyclic analog of 2-methyl-4-arylidene. Synthesis of $[\mu\text{-SiMe}_2(\eta^5\text{-2-Me-4-(2,4-Me}_2\text{C}_4\text{H}_2\text{N)Ind})_2]\text{ZrCl}_2$ By: Ivchenko, P. V.; et al Russian Chemical Bulletin (2010), 58(3), 589-593.
1.1 Solvents: Acetic acid; 1 h, reflux		
Experimental Protocols		

Scheme 35 (1 Reaction)

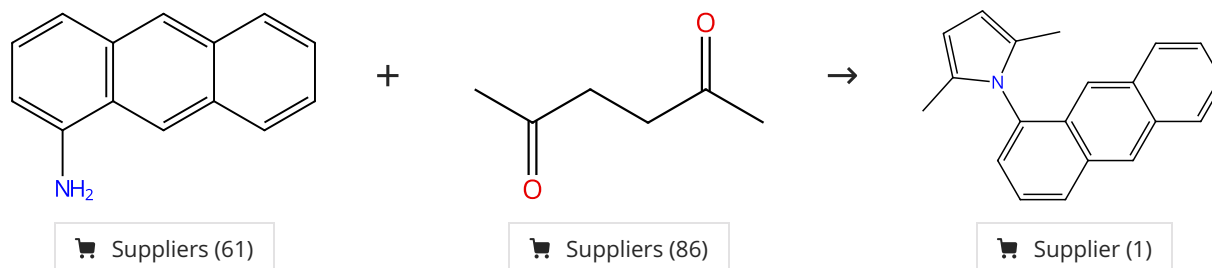
Steps: 1 Yield: 98%



31-309-CAS-11011412	Steps: 1 Yield: 98%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach By: Devi, Aarti; et al Synthetic Communications (2012), 42(10), 1480-1488.
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 36 min, rt		
Experimental Protocols		

Scheme 36 (1 Reaction)

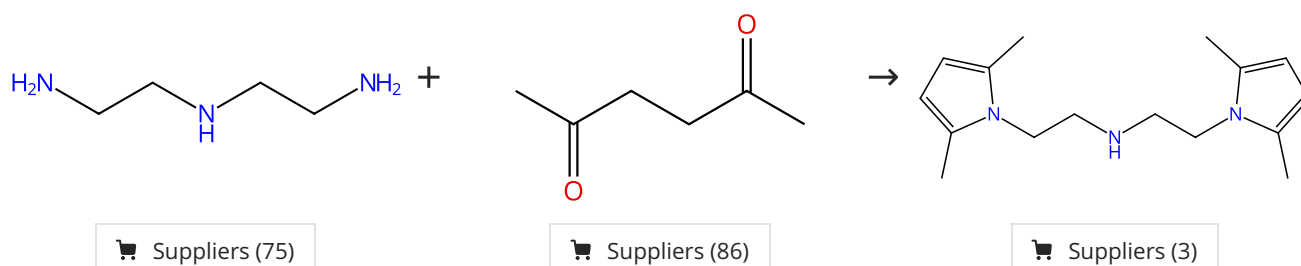
Steps: 1 Yield: 98%



31-309-CAS-3595812	Steps: 1 Yield: 98%	A straightforward highly efficient Paal-Knorr synthesis of pyrroles By: Banik, Bimal K.; et al Tetrahedron Letters (2005), 46(15), 2643-2645.
1.1 Catalysts: Bismuth nitrate		
Solvents: Dichloromethane; 19 h, rt		

Scheme 37 (2 Reactions)

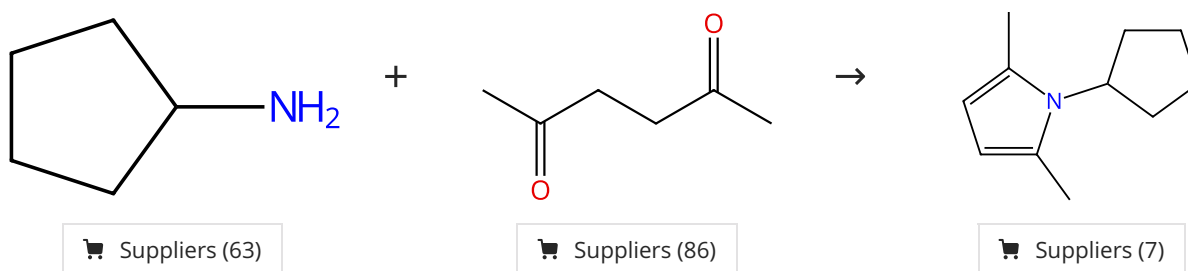
Steps: 1 Yield: 98%



31-309-CAS-7912058 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, Chlorosulfonic acid (silica-supported); 3 min, rt Experimental Protocols	Silica sulfuric acid (SSA) as a solid acid heterogeneous catalyst for one-pot synthesis of substituted pyrroles under solvent-free conditions at room temperature By: Veisi, Hojat Tetrahedron Letters (2010), 51(16), 2109-2114.
31-309-CAS-1509949 Steps: 1 Yield: 98% 1.1 Catalysts: <i>N</i> ¹ , <i>N</i> ¹ , <i>N</i> ³ , <i>N</i> ³ -Tetrachloro-1,3-benzenedisulfonamide Solvents: Acetonitrile; 5 min, rt	One-pot synthesis of substituted pyrroles with <i>N,N,N',N'</i>-tetrachlorobenzene-1,3-disulfonamide and <i>N,N'</i>-diiodo-<i>N,N'</i>-1,2-ethanediylbis(<i>p</i>-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.

Scheme 38 (2 Reactions)

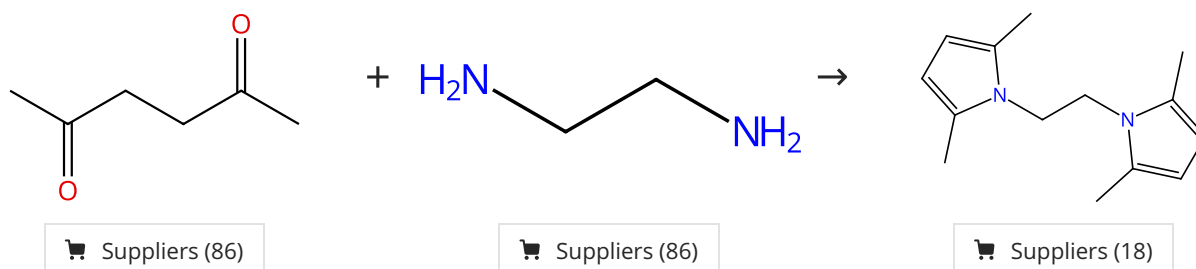
Steps: 1 Yield: 98%



31-309-CAS-3627207 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, Chlorosulfonic acid (silica-supported); 2 min, rt Experimental Protocols	Silica sulfuric acid (SSA) as a solid acid heterogeneous catalyst for one-pot synthesis of substituted pyrroles under solvent-free conditions at room temperature By: Veisi, Hojat Tetrahedron Letters (2010), 51(16), 2109-2114.
31-309-CAS-5463043 Steps: 1 Yield: 98% 1.1 Catalysts: <i>N</i> ¹ , <i>N</i> ¹ , <i>N</i> ³ , <i>N</i> ³ -Tetrachloro-1,3-benzenedisulfonamide Solvents: Acetonitrile; 2 min, rt	One-pot synthesis of substituted pyrroles with <i>N,N,N',N'</i>-tetrachlorobenzene-1,3-disulfonamide and <i>N,N'</i>-diiodo-<i>N,N'</i>-1,2-ethanediylbis(<i>p</i>-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.

Scheme 39 (2 Reactions)

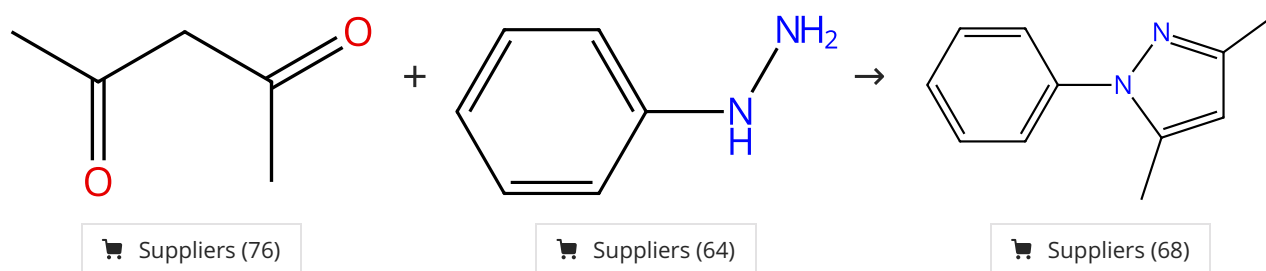
Steps: 1 Yield: 98%



31-309-CAS-5780379 Steps: 1 Yield: 98% 1.1 Catalysts: Silica, Chlorosulfonic acid (silica-supported); 5 min, rt Experimental Protocols	Silica sulfuric acid (SSA) as a solid acid heterogeneous catalyst for one-pot synthesis of substituted pyrroles under solvent-free conditions at room temperature By: Veisi, Hojat Tetrahedron Letters (2010), 51(16), 2109-2114.
31-309-CAS-7600620 Steps: 1 Yield: 98% 1.1 Catalysts: <i>N</i> ¹ , <i>N</i> ¹ , <i>N</i> ³ , <i>N</i> ³ -Tetrachloro-1,3-benzenedisulfonamide Solvents: Acetonitrile; 1 min, rt	One-pot synthesis of substituted pyrroles with <i>N,N,N',N'</i>-tetrachlorobenzene-1,3-disulfonamide and <i>N,N'</i>-diiodo-<i>N,N'</i>-1,2-ethanediylbis(<i>p</i>-toluenesulfonamide) as novel catalytic reagents By: Ghorbani-Vaghei, Ramin; et al South African Journal of Chemistry (2009), 62, 33-38.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 97%



31-309-CAS-734645 Steps: 1 Yield: 97% 1.1 Catalysts: Glutathione (supported on iron oxide) Solvents: Water; 20 min, 140 °C Experimental Protocols	Magnetic nanoparticle-supported glutathione: a conceptually sustainable organocatalyst By: Polshettiwar, Vivek; et al Chemical Communications (Cambridge, United Kingdom) (2009), (14), 1837-1839.
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